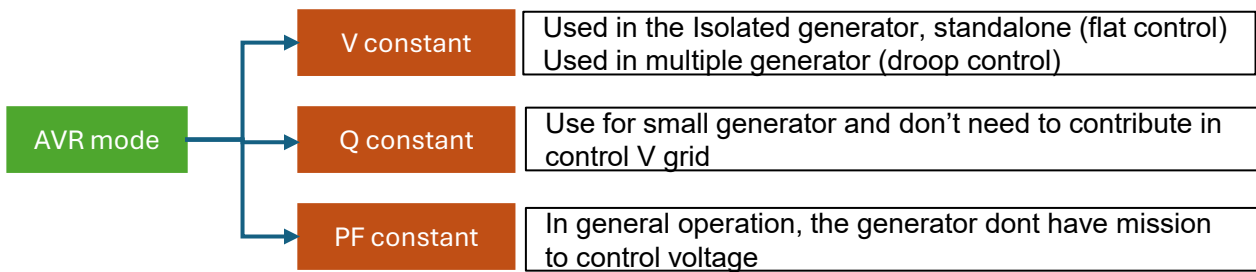


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AVR MODE AND COORDINATION



FLAT CONTROL

In this control mode, the generator produce/absorb the maximum Q the system need to keep the V constant.

However it cant operate in parralel with other generator due to the different in time respone and size of generator. The unequal reactive power sharing is the result of this coordinate control (1 máy dành hết phần phát Q của các máy còn lại)

DROOP CONTROL

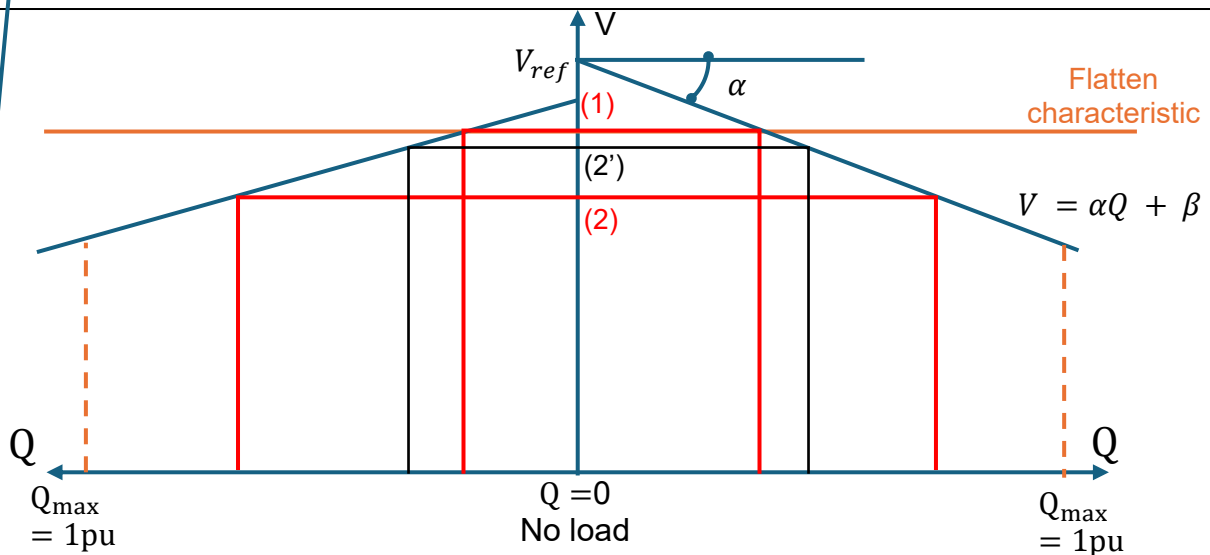
In this control mode, the amount of Q produce can control by

- The droop setting α form 4% to 6%, meaning the full 4% voltage drop is needed to drive reactive power (Q) from 0 to 100%. The larger slope the smaller Q need to produce (It is the natural setting of the AVR and **cant be changed**)
- The setting value of voltage: V_{ref} = no load voltage. If want to decrease the Q => decrease V_{ref}

The droof control will limit the maximum reactive power(Q) generator can produce although have different time respone.

Advantage: can control 2 different type of generator

Disadvantage: the voltage after setting operate with a little lower voltage (2') compared to initial voltage (1).



When voltage drop from (1) to (2) the new operating point set for 2 generator (the AVR increase the Q limit to support V)

When to generator produce more Q, the V increase to (1) however the operating point don't allow to produce more Q => V decrease

The new operating point will stop at (2') between (1) and (2) to balancing the Q produce and voltage.

FREQUENCY

Represent the different between the active power (P) of generator and the Load consumption
Not locally, it affect the hold system

$$2H \cdot \frac{df}{dt} = P_{gen} - P_{load} = \Delta P$$

- H: inertia of system depend on the number of motor due to the ability to store the energy in the kinetic form of motor. => the more heavy of system, the more stable ones.
- $\frac{df}{dt}$: ROCOF rate of changing on frequency

LOW INERTIA H

Appear in small system with few number of motor
Inverter – based system (PV, Wind, ...).
Due to the speed on/off and high sensitive to overvoltage, current of electronic devices.
So it can trip immediately when the fault happen.
=> No energy convert to slow down fault effects.

WHEN LOSS THE GENERATOR

$$P_G \downarrow \Rightarrow \Delta P < 0 \Rightarrow \frac{df}{dt} < 0 \Rightarrow f \downarrow$$

=> **The speed of rotor** ↓

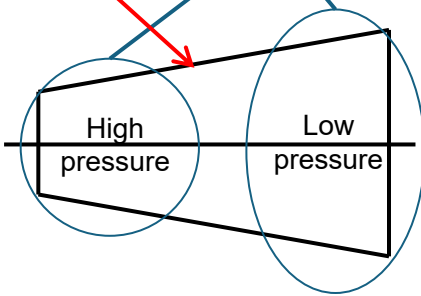
- How to respond to the decrease of frequency?
- What is the effect of f drop down?

1

Mechanical problem



Same energy due to the area



Turbine Blade Resonance

In low frequency, it can be the same as resonance frequency of turbine with can destroy the mechanic turbine

Over fluxing in the Winding of transformer of generator

$$E = 4.44 \cdot f \cdot \Phi \cdot K$$

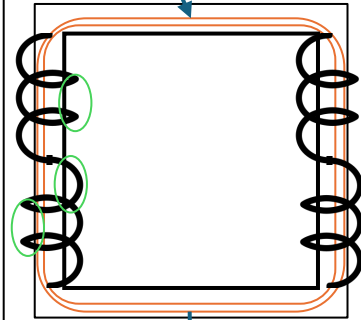
The flux in generator (Φ) is propotional to $\frac{V}{f}$
=> The flux in the core reach the limit and the retain flux overflow
=> Lead to thermal effect to the suporting components which don't design to conduct the flux => damage equipment

2

Frequency Avalanche

From sigle fault to black out

1. F decrease due to faults
2. Pump/fan in generato decrease the power due to f decrease
3. The boiler is forced to reduce output or trip to ensure safe operation.
4. The system generation power P_G becomes further deficient.
5. Frequency declines even more severely
→ leading to complete system separation (total blackout).



3

Voltage collapse

From sigle fault to black out

F decrease due to faults, and the core is overload and lead to the saturation
=> causing the Magnetizing Reactance (X_m) to drop sharply
=> The system acts like a short circuit for the magnetizing branch, drawing a massive Magnetizing Current
=> This current absorbs a large amount of Reactive Power (Q) from the grid
=> U decrease
=> Double contingency

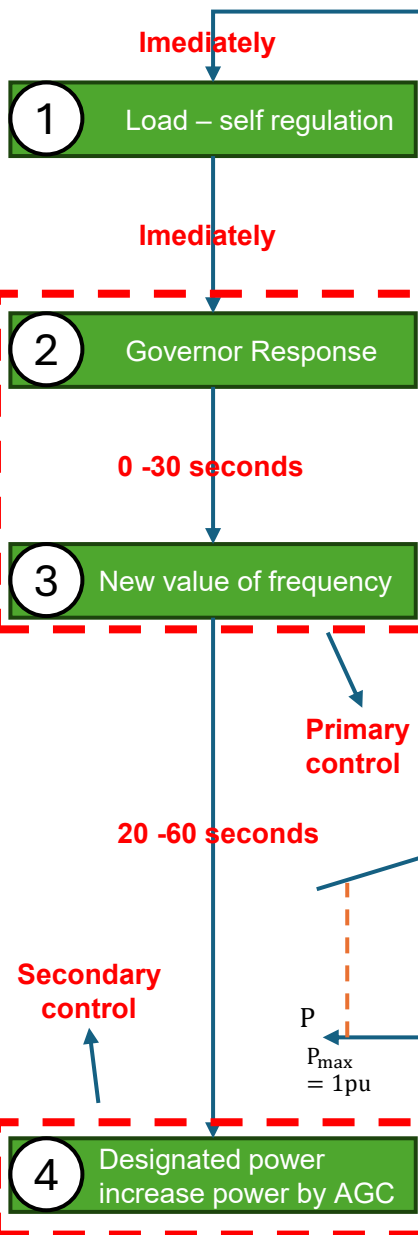
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FREQUENCY

WHEN LOSS THE GENERATOR

$$P_G \downarrow \Rightarrow \Delta P < 0 \Rightarrow \frac{df}{dt} < 0 \Rightarrow f \downarrow \Rightarrow \text{The speed of rotor} \downarrow$$

=> How to respond to the decrease of frequency?



Load damping (Đặc tính tự điều chỉnh của phụ tải)

Power consumption of motor is proportional to frequency and stored as kinetic energy ($P \propto f^3$ or $P \propto f^2$)

$$f \downarrow \Rightarrow \text{The speed of rotor} \downarrow \Rightarrow P_{load} \downarrow \Rightarrow \Delta P \downarrow$$

The kinetic energy transform back to electric to slow down the

decrease of frequency by damping factor D (1% -3%)

Mean when the f decrease 1% the P_{load} decrease 1% - 3%

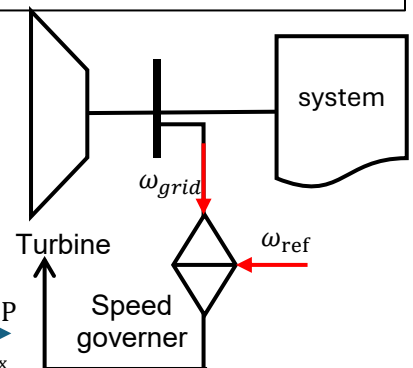
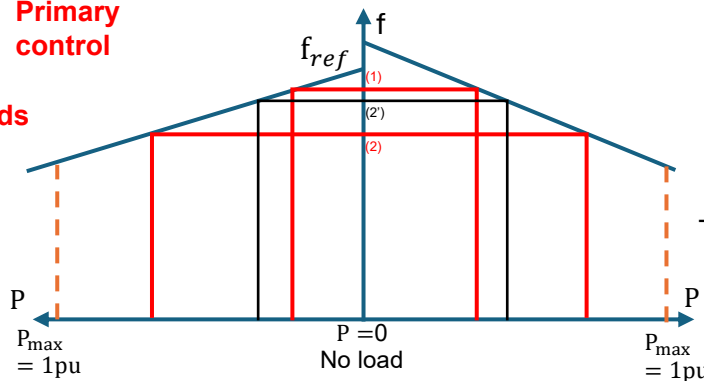
The speed governor see the speed decrease -> increase the power into turbine (Water, coal, gas ...)

Spinning reserve (the power ready to produce in the synchroized generator) is used.

Need time to ramp up and ramp down of turbine (3-10seconds)

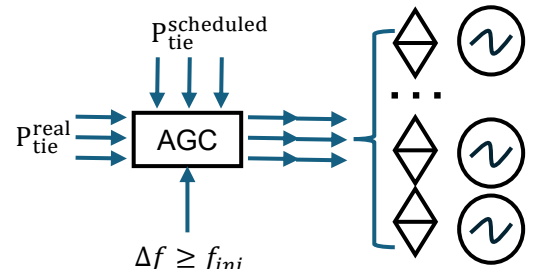
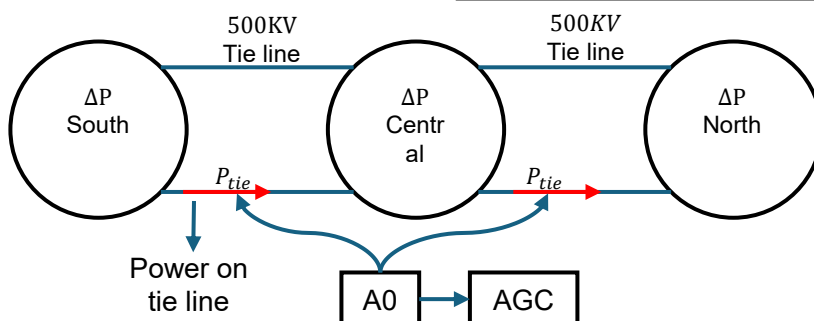
The coordinate of speed governor of multiple generator using Droop setting => If the generator can handle the fault, the new operation point still lower than initial ($(2') < (1)$)

The dispatcher change the f_{ref} to keep the voltage in range $\pm 0.2\text{Hz}$



AUTOMATIC GOVERNOR CONTROL AGC

- Compare P_{tie} with the schedule to indicate the fault location
- And order nearest designated power plant to produce more P
- Economic distributed after control



Automation in Power system

FREQUENCY

4 Designated power increase power by AGC

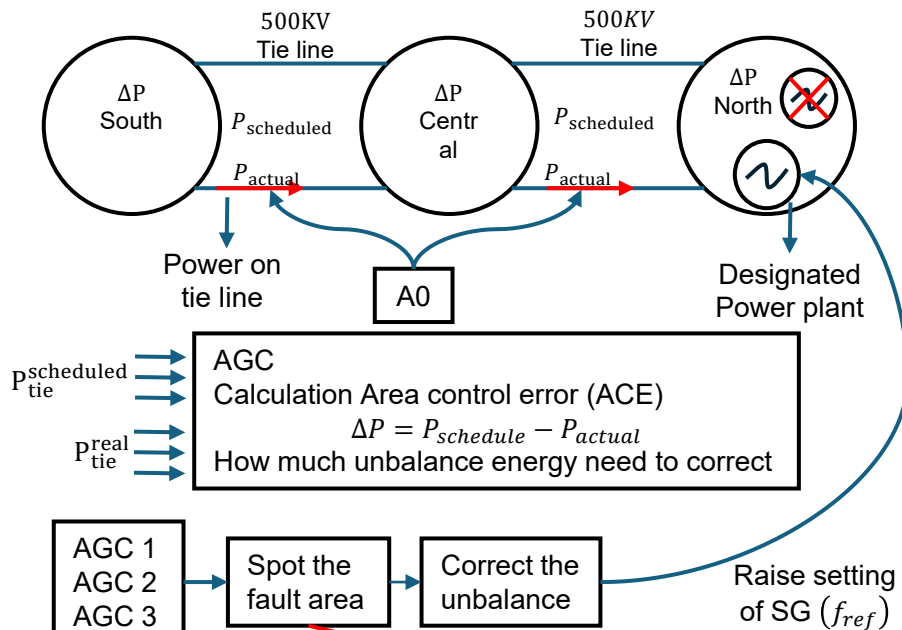
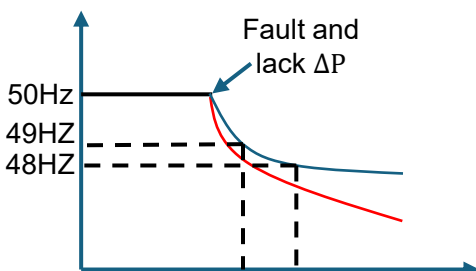
If (4) process can handle the fault and stop frequency drop:

- the frequency can return to initial value (50Hz)
- Designated Power plant handle all the ΔP
- Others: return to initial operation point

Else (4) process fail to handle the fault and frequency drop to threshold:

5 Economic distribution Recovery reserve power

6 The last defense Load Shedding



- Planned: $P_{schedule}^1 = 0$ MW
 $P_{schedule}^2 = 0$ MW
 - Actual $P_{actual}^1 = 100$ MW
 $P_{actual}^2 = 180$ MW
- ACE1 = $0 - (-100) = 100$ MW
ACE2 = $0 - (-180 + 100) = 80$ MW
ACE3 = $0 - (+180) = -180$
- More Energy
Lack Energy

Falut at the North

- ACE > 0 => healthy area
 - ACE < 0 => fault area
- Actual data:
- $$B \cdot \Delta f + \Delta P = 0$$
- With B is the true value of ACE
- if B too high it need the adjust pass over 0 point => need more time to react
 - If B too low, take more time to Δf convergent to f_{new}
- => Need to choose B carefully

- NLDC (A0) order other power plant to increase output
 - Designated power plant output decrease
- => Recover the f controlled ability (Spinning reserve) and ready cho the next fault

Using UC to choose the plant to increase

Condition to active:

- ΔP very high and (4) cant handle
- The drop speed of frequency. (ROCOF) extreme high which SG and AGC cant adapt.
- The frequency drop lower than the safety threshold.

Under frequency load shedding

- When frequency reach the threshold, ULCS trip load P1
- ROCOF decrease and the frequency is more flatten
- Reach the new value $f_{min} > 47.5$ Hz and start to recover

6

The last defense Load Shedding

Reclose Load

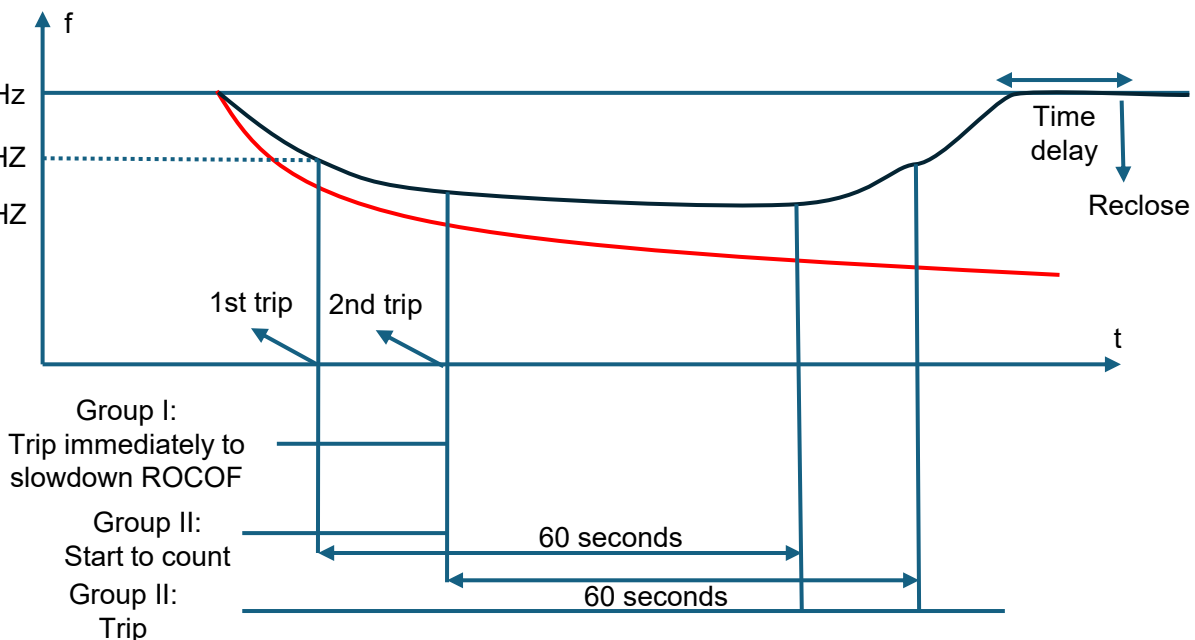
- Recovery load fast
- Avoid misoperation

Condition:

- Frequency recovery to 49.5 and 50Hz
- Time delay: 10-20s
- Only reclose 1 time only, and lockout if fail
- Don't close too many loads in the same time due to the effect of **inrush current** which can destroy the frequency

Group divide:

- **Group 1(Arresting group):** Slow down immediately the frequency drop (ROCOF↓)
 - Active from many level from 49.2Hz to 47.5Hz
 - With the low H system, using the $df/dt > 1.5\text{Hz/s}$ to trip the load
- **Group 2(Resoring group):** Restore the frequency to 50Hz using cascading delay
- **Group 3(supplemental group):** Back up protection for the serious case, active when ROCOF too high and V drop very low



ROCOF relay 81

- Principle: Compare $f_{meas} < f_{set}$, a duration $t > t_{delay}$, then TRIP
- Can't see the serious of problem
- Prone to **Overshedding**

Block ROCOF relay 81R

- Voltage Blocking $U < 0.8U_{rated} \Rightarrow$ closed to avoid trip while sc.
- Sliding window: Cal ROCOF in average window 100-500ms
- Logic to cut $df/dt > \text{threshold}$ and $f < \text{threshold}$

ROCOF relay 81R

- Principle: $df/dt \Rightarrow \Delta P$
 - Can see the serious of problem
- Phase Jump: when close capacitor or short circuit the **phase jump a distance** and create a **virtual frequency** $2\pi f = \omega = \frac{d\theta}{dt} \rightarrow \infty \Rightarrow$ trip and misoperation
- Zero crossing fail: The relay measure the frequency by **zero crossing points** $\Rightarrow$ if the voltage move by harmonic or short circuit $\Rightarrow$ wrong f calculation.